MATH 112 — Practice Exam 2A

Open notes **Suggested time: 80 minutes** **Total: 100 points**

Name: _____

Instructions. Attempt all seven problems. Show your work and justify your answers except where explicitly stated otherwise.

Problem 1. True or False (16 points)

For each statement, circle **T** or **F**. Each part is worth **2 points**. **No justification is required.**

(a) Every lower bounded set in $\mathbb{R}$ has an infimum in $\mathbb{R}$. T F

(b) Every upper bounded set in $\mathbb{Q}$ has a supremum in $\mathbb{Q}$. T F

(c) If there exist x, y such that for all $\varepsilon > 0, |x - y| < \varepsilon$, then $x = y$. T F

(d) For any sequence (s_n), the sequence $(\frac{s_n}{n})$ converges. T F

(e) If a set A has a maximum, then it has a supremum and $\sup(A) = \max(A)$. T F

(f) If (a_n) is a sequence of integers, then (a_n) converges if and only if it is eventually constant. T F

(g) For every complex number $z \in \mathbb{C}, z^2 = z\bar{z}$. T F

(h) If $\Re(z) = 0$, then $|z| = \Im(z)$. T F

Problem 2 (15 points)

Find the supremum, infimum, maximum, and minimum, if they exist, of the following *subsets of* $\mathbb{R}$. Justify your answers.

(a) $A = \{x \in \mathbb{R} : x > 0,\ 1 < x^2 \leq 9\}$. (5 points)

(b) $B = \left\{ \dfrac{2n+1}{n} : n \in \mathbb{N} \right\}$. (5 points)

(c) $C = \left\{ (-1)^n + \dfrac{1}{n+1} : n \in \mathbb{N} \right\}$. (5 points)

Problem 3 (14 points)

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above.

(a) Suppose $\sup A < \sup B$. Show that there exists $b \in B$ such that $a < b$ for every $a \in A$. (7 points)

(b) Suppose there is $c > 0$ such that for every $a \in A$ there is $b \in B$ satisfying $a \leq b + c$. Prove $\sup A \leq \sup B + c$. (7 points)

Special-angle reference for Problems 4–5. You may use the following values.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
$\sin\theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\cos\theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0	$-1/2$	$-\sqrt{2}/2$	$-\sqrt{3}/2$	-1

Problem 4 (16 points)

(a) Write $\dfrac{2+i}{1-i}$ in the form $a+bi$. (4 points)

(b) Let $z=-\sqrt{3}+i$. Find $|z|$, an argument $\theta\in[0,2\pi)$, and write z in polar form. (4 points)

(c) Use de Moivre's formula to express z^6 from part (b) in the form $a+bi$. (4 points)

(d) Describe geometrically what multiplication by i does to a nonzero complex number. (4 points)

Problem 5 (12 points)

Consider

$$w=-\frac{1}{2}+\frac{\sqrt{3}}{2}i.$$

(a) Express w in trigonometric form. (4 points)

(b) Find *one* complex number $z=a+bi$ such that $z^4=w$. Specify $a,b\in\mathbb{R}$. (4 points)

(c) Without calculating the other roots explicitly, explain how many distinct fourth roots w has and describe their geometric arrangement in the complex plane. (4 points)

Use the special-angle reference table above as needed.

Problem 6 (12 points)

Throughout, give geometric descriptions where possible.

(a) In $\mathbb{C}$ with $d(z,w)=|z-w|$, describe $B(1+i,2)$. Determine whether $3+i$ and $2+2i$ belong to this ball. (5 points)

(b) Describe geometrically $\{z\in\mathbb{C}:|z-1|<|z+1|\}$. (3 points)

(c) Define $\rho(z,w)=2|z-w|$. Verify that ρ is a metric on $\mathbb{C}$ and describe $B_\rho(0,2)$. (4 points)

Problem 7 (15 points)

(a) Using the ε–$n_0(\varepsilon)$ definition, prove that $a_n=\dfrac{n+2}{n+4}\to 1$. Give an explicit valid choice of $n_0(\varepsilon)$. (5 points)

(b) Use the Squeeze Theorem to prove that

$$\frac{\sin(n^2)+\cos n}{n}\longrightarrow 0.$$

(5 points)

(c) Let (a_n) be a sequence of real numbers such that $a_n \to 3$. Determine whether the sequence $b_n = (-1)^n + a_n$ converges. Justify your answer using an appropriate limit theorem. (5 points)